



DATA VISUALIZATION ASSIGNMENT 2

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1. Introduction

This report presents a detailed walkthrough of the Power BI dashboard developed for Assignment 2 of the Data Visualization course. The dashboard is built around a dataset of drone strike incidents, focusing on temporal trends, geographic distribution, and the human impact of these events across provinces and cities in Pakistan — particularly the Federally Administered Tribal Areas (FATA) and Khyber Pakhtunkhwa (KPK) regions.

The report explains every chart and visual element included in the dashboard: what each one shows, what data it uses, and how it was created in Power BI. The dashboard consists of three report pages, each designed to answer a specific set of analytical questions.

1.1 Dashboard Summary

3 Report Pages	38 Total Visuals	11 KPI Cards	15 Measures / Fields
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1.2 Report Pages Overview

Page	Title	No. of Visuals	Focus Area
1	Attack Timeline	18	Temporal trends in attacks and casualties
2	Geographic Hotspots	9	Location-based attack concentration
3	Human Impact Analysis	11	Civilian vs. militant casualties

2. Dataset and Calculated Measures

The dataset underpinning this dashboard contains records of drone strike incidents. Each row in the dataset represents a single strike event and includes fields such as the year, province, city, GPS coordinates, number of civilian deaths, militant deaths, and number of injuries. In addition, a set of DAX (Data Analysis Expressions) calculated measures were created to derive higher-level KPIs from the raw data.

2.1 Data Fields (Dimensions)

Measure / Field	Category	Description
Province	Dimension	Categorical field identifying which province the strike occurred in (e.g., FATA, KPK, Balochistan).
City	Dimension	City or locality within the province where the strike was recorded.
Year	Dimension	The year the drone strike took place, used for time-series analysis.
Latitude	Geographic	GPS latitude coordinate of the strike location, used in the map visual.
Longitude	Geographic	GPS longitude coordinate of the strike location, used in the map visual.

2.2 Calculated Measures (DAX)

The following measures were created using DAX formulas to aggregate and derive KPIs displayed in card visuals and charts:

Measure / Field	Category	Description
Total Attacks	Count KPI	COUNT or COUNTROWS of the strike events table. Gives the total number of recorded drone strikes.
Total Deaths	Sum KPI	SUM of all fatalities (civilian + militant) across all strikes.
Total Injuries	Sum KPI	SUM of the injuries column, counting all persons injured across all incidents.

Total Civilian Deaths	Human Impact	SUM of the civilian deaths field only, separating non-combatant fatalities.
Total Militant Deaths	Human Impact	SUM of the militant deaths field only, counting combatant fatalities.
Avg Deaths Per Attack	Ratio KPI	DIVIDE([Total Deaths], [Total Attacks]) — average lethality per single strike event.
Civilian Casualty %	Ratio KPI	DIVIDE([Total Civilian Deaths], [Total Deaths]) expressed as a percentage, showing what proportion of all deaths were civilians.
FATA Pct	Geographic	DIVIDE of FATA-region deaths or attacks to total, showing FATA's share of overall impact.
Most Affected Province	Text KPI	Calculated using TOPN or MAXX to return the province name with the highest attack count.
Most Affected City	Text KPI	Similar to Most Affected Province but at the city level.
Total Locations	Count KPI	DISTINCTCOUNT of city or coordinate pairs to show how many unique strike locations exist.
Highest Death Province	Text KPI	Province name where the SUM of Total Deaths is highest.

3. Page 1 — Attack Timeline

The first page of the dashboard, Attack Timeline, focuses on the temporal dimension of the data. It answers the question: how have drone strike frequency and casualties changed over time? It contains 18 visual elements including 5 KPI cards, 3 charts, 3 interactive slicers, and 4 branded images.

3.1 KPI Cards

Five KPI cards are placed prominently at the top of the page to give an immediate summary of the most critical numbers at a glance.

#	Visual Type	Title / Name	What It Shows
1	KPI Card	Total Attacks	Displays the total number of drone strike events in the entire dataset (or filtered by slicers). Built by inserting a Card visual and placing the Total Attacks measure in the Fields well.
2	KPI Card	Total Deaths	Shows the aggregate death toll across all strikes. Built with the Total Deaths measure in a Card visual. This is the most impactful headline number.
3	KPI Card	Total Injuries	Displays the count of all persons injured (not killed) across all strikes. Uses the Total Injuries measure in a Card visual.
4	KPI Card	Avg Deaths Per Attack	Shows the average lethality of a single strike, calculated as Total Deaths divided by Total Attacks (DAX)

			DIVIDE). Helps the viewer understand the average severity of each event.
5	KPI Card	Civilian Casualty %	Displays the civilian casualty rate as a percentage. This is the most sensitive KPI, showing what share of all deaths were non-combatants. Built with the Civilian Casualty % DAX measure.

3.2 Line Chart — Drone Attacks Over Year

What It Shows

This chart plots the number of drone strikes per year on the Y-axis against year on the X-axis. It reveals the trend in attack frequency — whether strikes increased, peaked, or declined over the study period. Viewers can instantly see which years had the most activity.

How It Was Made

- Visual type: Line Chart, inserted from the Visualizations pane.
- X-axis: Year field dragged to the Axis well.
- Y-axis: Total Attacks measure dragged to the Values well.
- Formatting: A meaningful title 'Drone Attacks Over Year' was added. The line color and marker style were customized in the Format pane.
- The chart automatically aggregates the Total Attacks measure by each year.

3.3 Area Chart — Deaths and Injuries Over Time

What It Shows

This area chart overlays two series — Total Deaths and Total Injuries — over time. The filled areas allow easy visual comparison of the magnitude of each category year by year. It highlights years where injuries and deaths were unusually high relative to the number of strikes.

How It Was Made

- Visual type: Area Chart from the Visualizations pane.
- X-axis: Year field.
- Y-axis (Values): Both Total Deaths and Total Injuries measures added to the Values well, creating dual series.
- Each series automatically receives a different color. Transparency was adjusted in the Format pane to allow both areas to be visible where they overlap.

3.4 Clustered Column Chart — Civilian vs. Militant Deaths Over Time

What It Shows

This chart directly compares civilian and militant death counts side by side for each year. It answers the critical question: in any given year, what proportion of deaths were civilians versus combatants? Tall civilian bars relative to militant bars indicate particularly high collateral impact.

How It Was Made

- Visual type: Clustered Column Chart.
- X-axis: Year field.
- Y-axis (Values): Total Civilian Deaths and Total Militant Deaths measures both added to the Values well, creating two grouped columns per year.
- Title set to 'CIVILIAN VS MILITANT DEATH OVER TIME'.
- Colors for each series (e.g., red for civilian, grey for militant) were set in the Data Colors section of Format pane.

3.5 Stacked Area Chart — Province-wise Attack Trend

What It Shows

This stacked area chart shows how the total volume of attacks is distributed across provinces over time. Each colored band represents one province's contribution to the total for that year. Wider bands in a given year mean more strikes in that province relative to others. This visual is effective for understanding both the overall growth in attacks and the shifting geographic focus.

How It Was Made

- Visual type: Stacked Area Chart.
- X-axis: Year field.
- Y-axis (Values): Total Attacks measure.
- Legend: Province field dragged to the Legend well — this splits the area into one colored band per province.
- Power BI automatically stacks and color-codes each province's contribution.

3.6 Interactive Slicers (Filters)

Three slicer visuals are placed on this page to allow the user to filter all charts dynamically:

- Province Slicer — allows filtering by province. All charts on the page update to show only that province's data.
- Year Slicer — a range slider or dropdown for filtering by year. Useful for zooming into specific time periods.
- City Slicer — allows filtering down to a specific city within the selected province.

How they were made: Each slicer was created by selecting the Slicer visual type and dragging the respective field (Province, Year, City) into the Field well. Slicer type (e.g., dropdown, list, or slider) was set in the Format pane under Slicer Settings.

4. Page 2 — Geographic Hotspots

The second page shifts focus from time to space. It answers the question: where did strikes occur, and which locations were most severely impacted? It contains 9 visuals: 4 KPI cards, a geographic map, a bar chart, a treemap, a pivot table, and a text annotation box.

4.1 KPI Cards

#	Visual Type	Title / Name	What It Shows
1	KPI Card	Most Affected Province	Displays the name of the single province with the most recorded strikes. Built using the Most Affected Province DAX text measure in a Card visual.
2	KPI Card	Most Affected City	Displays the city with the highest strike count. Uses the Most Affected City DAX measure.
3	KPI Card	Total Locations	Shows how many distinct geographic locations (unique cities or coordinates) appear in the dataset, using the Total Locations DISTINCTCOUNT measure.
4	KPI Card	Highest Death Province	Displays the province with the largest cumulative death toll — which may differ from the most attacked province if some regions suffer more fatalities per strike.

4.2 Geographic Map

What It Shows

The map visual plots each strike location as a bubble on a real geographic map of Pakistan. The size or color of each bubble can represent the number of strikes or total deaths at that location. This gives an immediate visual sense of where attacks are concentrated — typically in the tribal belt along the Afghan border.

How It Was Made

- Visual type: Map (Bing Maps-backed built-in Power BI visual).
- Location: Latitude field added to the Latitude well, Longitude field added to the Longitude well. This places exact coordinate points on the map.
- Legend: Province field added to the Legend well to color-code bubbles by province.
- Size: Total Deaths or Total Attacks measure placed in the Size well so that locations with more impact show larger bubbles.
- In Format pane: map style, bubble transparency, and zoom level were adjusted for clarity.

4.3 Bar Chart — Attacks by City

What It Shows

This horizontal bar chart ranks cities by the number of strikes or deaths they recorded. The longest bar is the most affected city. It provides a precise ranked view of location impact that complements the map's spatial perspective.

How It Was Made

- Visual type: Clustered Bar Chart (horizontal).
- Y-axis: City field.
- X-axis (Values): Total Attacks or Total Deaths measure.
- Sorted descending by value so the most affected cities appear at the top.
- Data labels were enabled in Format pane to show exact values on each bar.

4.4 Treemap — City-level Proportional View

What It Shows

The treemap displays each city as a colored rectangle, with the rectangle's area proportional to that city's attack count or death toll. Larger rectangles immediately draw the eye to the most impacted locations. It complements the bar chart by emphasizing relative proportions rather than absolute ranked values.

How It Was Made

- Visual type: Treemap from Visualizations pane.
- Group: City field placed in the Group well — each rectangle represents one city.
- Values: Total Attacks (or Total Deaths) measure placed in the Values well — this sets the rectangle size.
- Colors were automatically assigned per city. A category label and data label were enabled to show city name and count inside each rectangle.

4.5 Pivot Table — Province Summary

What It Shows

The matrix/pivot table provides a structured tabular breakdown of key metrics by province. It shows, for each province: total attacks, total deaths, total injuries, and other relevant figures side by side in rows and columns. This is useful for viewers who prefer precise numbers over visual representations.

How It Was Made

- Visual type: Matrix (Power BI's pivot table visual).
- Rows: Province field.
- Values: Multiple measures placed in the Values well (Total Attacks, Total Deaths, Total Injuries, etc.).
- Row subtotals and grand totals enabled via Format pane under Row Subtotals.

5. Page 3 — Human Impact Analysis

The third page dives deep into the human consequences of drone strikes. It answers: how are deaths distributed between civilians and militants, which provinces and cities suffer most in terms of casualties, and what patterns emerge when we look at strike intensity against fatality scale? It contains 11 visuals: 5 KPI cards, a column chart, a donut chart, a scatter chart, a decomposition tree, and 2 text boxes.

5.1 KPI Cards

#	Visual Type	Title / Name	What It Shows
1	KPI Card	Civilian Deaths	Total count of civilian fatalities across all strikes. The most ethically significant KPI in the report.
2	KPI Card	Total Militant Deaths	Total count of combatant fatalities, for comparison with civilian deaths.
3	KPI Card	Total Injuries	Count of all injuries (regardless of civilian/militant status).
4	KPI Card	Civilian Casualty %	Repeats the civilian casualty rate from Page 1, now in the context of the full human impact page for easy reference.
5	KPI Card	FATA Pct	Percentage of total deaths or strikes attributable to the FATA region specifically, highlighting its disproportionate share of the overall impact.

5.2 Column Chart — Casualties by City

What It Shows

A vertical column chart showing total casualties (deaths or injuries) broken down by city. This provides a city-level ranked view of human impact, similar to the bar chart on Page 2 but focused on casualty counts rather than just attack frequency.

How It Was Made

- Visual type: Clustered Column Chart.
- X-axis: City field.
- Y-axis: Total Deaths (or Total Civilian Deaths) measure.
- Sorted descending so the hardest-hit cities appear first on the left.
- Data labels enabled. Color customized in Format pane.

5.3 Donut Chart — Proportional Deaths by Province

What It Shows

The donut chart shows each province's share of total deaths as a colored arc segment. The full circle represents 100% of all deaths. Larger arcs indicate provinces that account for a greater proportion of total fatalities. This is one of the clearest ways to communicate relative provincial impact at a glance.

How It Was Made

- Visual type: Donut Chart.
- Legend: Province field placed in the Legend well — each province becomes a colored slice.
- Values: Total Deaths measure placed in the Values well — this sets each slice's size proportionally.
- Data labels showing percentage or value were enabled.
- In Format pane: inner radius was set to create the donut hole, colors were assigned per province, and a title was added.

5.4 Scatter Chart — Strike Intensity vs. Total Fatality Scale

What It Shows

This scatter chart plots each province (or data point) as a dot, with one axis representing strike intensity (e.g., number of attacks) and the other representing total fatality scale (e.g., total deaths). Dots in the upper-right quadrant represent provinces that experienced both many strikes and high deaths. Dots in the upper-left would indicate high deaths from fewer but more lethal strikes. This chart reveals whether more attacks necessarily means more deaths, or whether some provinces suffer disproportionate casualties per strike.

How It Was Made

- Visual type: Scatter Chart.
- X-axis: Total Civilian Deaths (or Total Attacks) measure.
- Y-axis: Total Deaths measure.
- Details: Province field placed in the Details well — each dot represents one province.
- Play Axis can optionally be added with Year to animate the scatter over time.
- Bubble size was left uniform or set to a third measure for additional dimensionality.

5.5 Decomposition Tree — Casualty Drill-Down

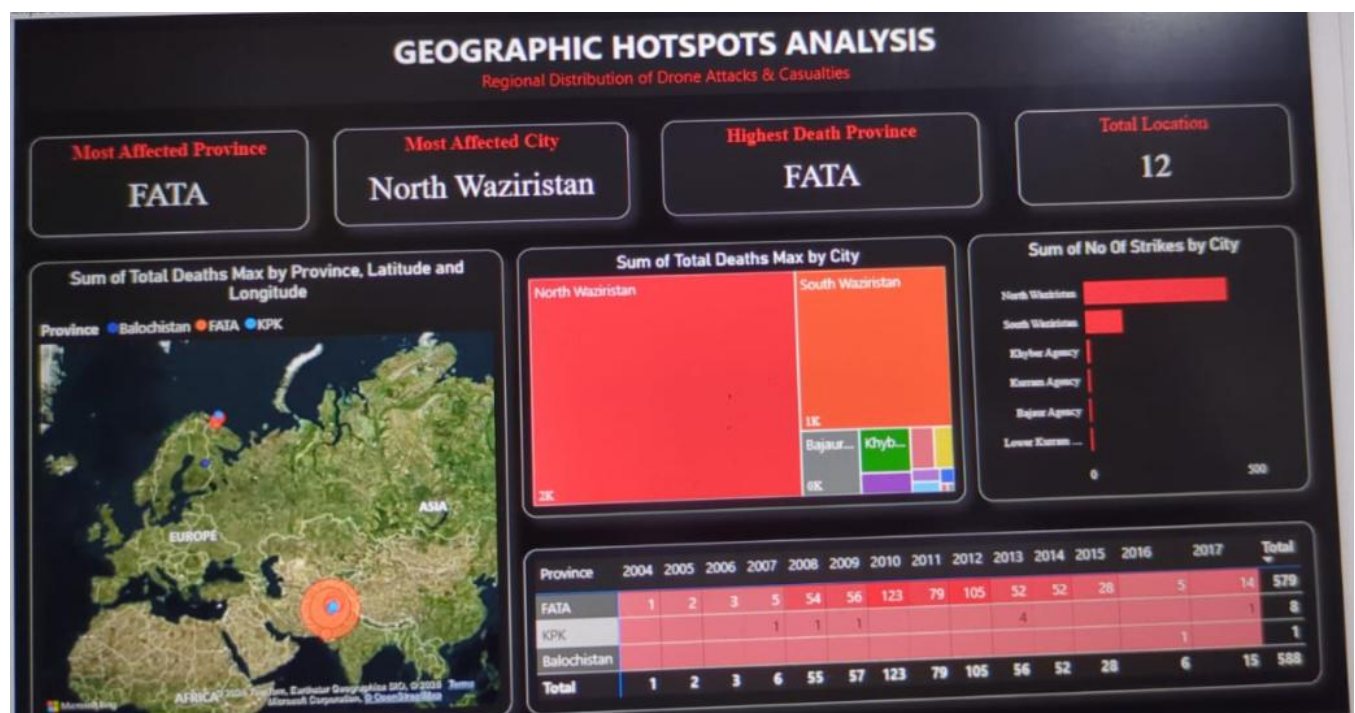
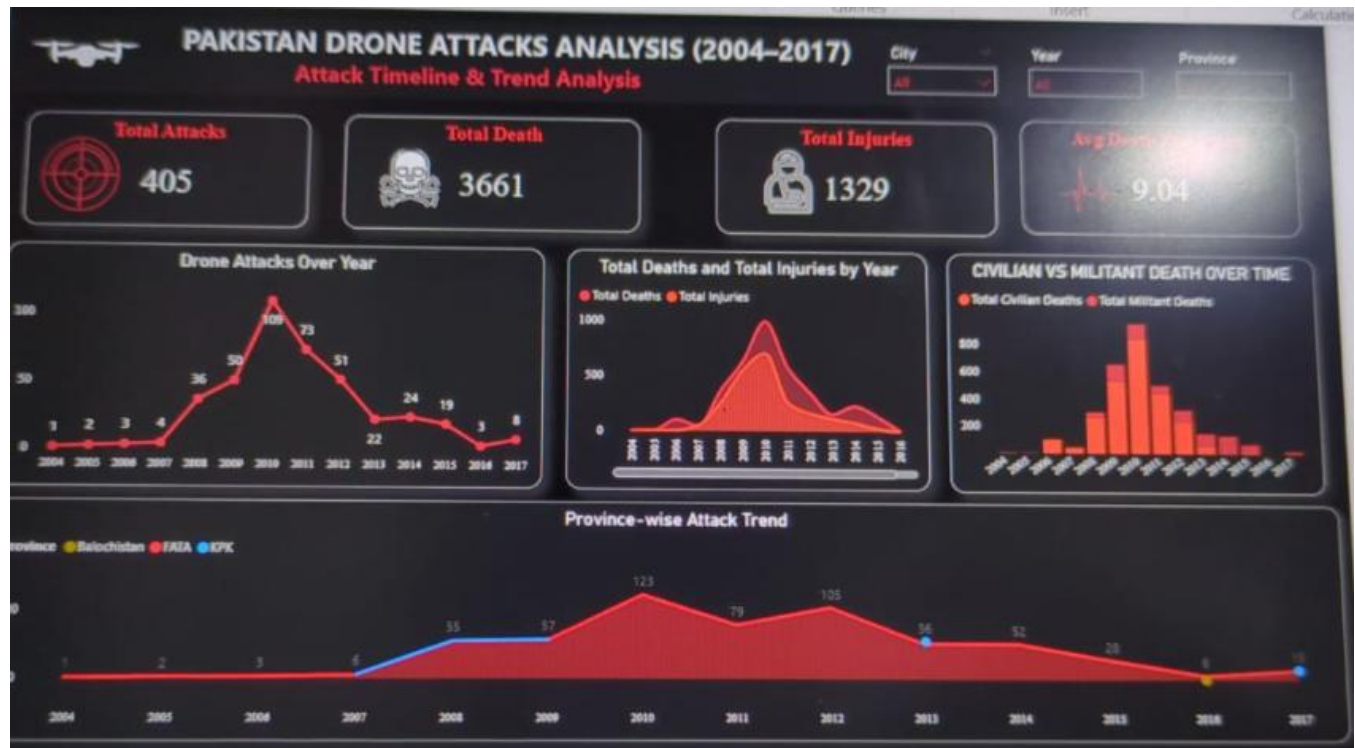
What It Shows

The decomposition tree is the most interactive visual on the entire dashboard. It allows the user to drill down from the top-level Total Deaths figure, breaking it down first by Province, then by City within that province, then by Year within that city. At each level, the tree shows which sub-category contributes most to the total. This is invaluable for root-cause analysis — for example: 'Most deaths come from FATA, specifically from Miranshah, and the worst year was 2010.'

How It Was Made

- Visual type: Decomposition Tree (AI-powered visual built into Power BI).
- Analyze: Total Deaths measure placed in the Analyze well — this is the value being decomposed.
- Explain By: Province, City, and Year fields all placed in the Explain By well in sequence.
- At runtime, the user clicks the '+' button on any node to expand into the next level of breakdown. Power BI can also auto-select the highest-value path (High Value AI split).
- Title set to 'Casualty Drill-Down: Province, City, and Year'.

Final Output:





6. Design Choices and Formatting

Intentional design decisions were made throughout the report to improve readability, visual consistency, and user experience.

6.1 Theme

- Power BI theme CY26SU02 (2026.03 cloud release) was applied as the base theme, providing a consistent color palette across all visuals.
- Five AI-generated (Google Gemini) images were embedded as visual assets on Page 1, giving the dashboard a professional, branded appearance relevant to the subject matter.

6.2 Interactivity

- Three slicers on Page 1 (Province, Year, City) enable cross-filtering — when the user selects a value in a slicer, all charts on the page automatically update to reflect that filter.
- The Decomposition Tree on Page 3 provides deep drill-down capability, allowing users to explore data hierarchically without needing to change the page.
- The Geographic Map on Page 2 supports pan and zoom for exploring specific areas.

6.3 Page Structure

- Each page follows a consistent layout: KPI cards at the top for instant insight, primary charts in the centre, and secondary/supporting charts below or to the side.
- Text boxes were used on pages 2 and 3 to add brief explanatory annotations, helping a non-expert viewer understand the context of what they are seeing.

6.4 Visual Hierarchy

- KPI cards are the first thing the eye sees on each page, anchoring the viewer with the most important numbers before they engage with the detail charts.
- The line chart and stacked area chart on Page 1 are the largest visuals, reflecting their importance as the primary analytical charts.
- Supporting charts (treemap, pivot table, donut) are sized smaller as supplementary perspectives.

7. Conclusion

This Power BI dashboard successfully addresses three key dimensions of the drone strike dataset: when attacks happened and how they trended over time (Page 1), where they were concentrated geographically (Page 2), and what their human impact was in terms of civilian versus militant casualties (Page 3).

The combination of KPI cards, trend charts, geographic maps, and advanced visuals such as the scatter plot and decomposition tree makes the dashboard both accessible to a general audience and powerful enough for detailed analytical exploration. The interactive slicers further enhance its utility by allowing targeted filtering by province, city, and year.

Each chart type was chosen deliberately: line and area charts for trends, bar and column charts for rankings, maps for spatial distribution, donut charts for proportional breakdowns, and the decomposition tree for hierarchical drill-down. Together they form a cohesive, multi-dimensional analytical story about the impact of drone strikes in Pakistan's conflict-affected regions.

7.1 Summary of All Visuals

Pg	Visual Type	Title	Data Used	Purpose
1	KPI Card	Total Attacks	Total Attacks measure	Headline count of all strikes
1	KPI Card	Total Deaths	Total Deaths measure	Headline fatality count
1	KPI Card	Total Injuries	Total Injuries measure	Count of non-fatal casualties
1	KPI Card	Avg Deaths Per Attack	Avg Deaths Per Attack	Lethality per strike
1	KPI Card	Civilian Casualty %	Civilian Casualty %	Non-combatant impact rate
1	Line Chart	Drone Attacks Over Year	Year, Total Attacks	Attack frequency trend
1	Area Chart	Deaths & Injuries Trend	Year, Deaths, Injuries	Casualty trend over time

1	Column Chart	Civilian vs Militant Deaths	Year, Civilian, Militant Deaths	Casualty type comparison
1	Stacked Area	Province-wise Attack Trend	Year, Province, Attacks	Provincial share over time
1	Slicer	Province Filter	Province	Cross-filter by province
1	Slicer	Year Filter	Year	Cross-filter by year
1	Slicer	City Filter	City	Cross-filter by city
2	KPI Card	Most Affected Province	Most Affected Province	Top province by strikes
2	KPI Card	Most Affected City	Most Affected City	Top city by strikes
2	KPI Card	Total Locations	Total Locations	Unique strike sites
2	KPI Card	Highest Death Province	Highest Death Province	Province with most deaths
2	Map	Geographic Strike Map	Lat, Long, Province	Spatial strike distribution
2	Bar Chart	Attacks by City	City, Total Attacks	City-level ranked strikes
2	Treemap	City Proportional View	City, Total Attacks	Proportional city impact
2	Matrix	Province Summary Table	Province, all measures	Tabular province breakdown
3	KPI Card	Civilian Deaths	Total Civilian Deaths	Civilian fatality count
3	KPI Card	Total Militant Deaths	Total Militant Deaths	Combatant fatality count
3	KPI Card	Total Injuries	Total Injuries	Injury count
3	KPI Card	Civilian Casualty %	Civilian Casualty %	Civilian death rate
3	KPI Card	FATA Pct	FATA Pct	FATA region share
3	Column Chart	Casualties by City	City, Total Deaths	City-level casualties
3	Donut Chart	Deaths by Province	Province, Total Deaths	Provincial death shares
3	Scatter Chart	Strike Intensity vs Fatality	Deaths, Province, Civ Deaths	Intensity vs lethality pattern